

Solar heating and its influence on mixing in ice-covered lakes

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SUMMARY. 1. The influence of solar heating on the formation of temperature and density profiles of ice-covered lakes is considered.

2. Mathematical models are derived to quantify the effect of solar heating on vertical stability and its dependence on critical parameters.

3. Solar heating is shown to be able to account for the thermal structure and mixing patterns of certain ice-covered lakes.

4. The importance of convective mixing is discussed in relation to the distribution of phytoplankton and nutrients.

Introduction

Ice-covered lakes present a special situation where the effect of the wind in causing mixing is greatly reduced. Nevertheless, many studies have shown that mixing can occur under ice in lakes in temperate and polar regions (e.g. Sokolnikov, 1960; Likens & Ragotzkie, 1965, 1966; Hawes, 1983). Patterns on the ice surface suggesting the presence of convection cells have also been observed, by Woodcock & Riley (1947) and by the present authors. As the density of water is greatest at 4°C and as salts in solution are frozen out during ice formation, density-driven water movements have been invoked to provide mechanisms for mixing under ice. These include (1) warming of water below the ice by solar radiation, which then convects downwards due to its higher density (Farmer, 1975; Petrov, 1984), (2) density currents flowing down the sides of lakes due to the release of stored heat from the sediments or geothermal heating (Welch &

Bergmann, 1985), (3) freezing out of salts as the ice layer thickens, forming a denser unstable layer below the ice (British Antarctic Survey, unpublished data). Mixing under ice may also be caused by turbulent shear flow from rivers or meltwater (Bergmann & Welch, 1985), or in large lakes by wind-induced movements of the ice sheet (Stewart, 1973). Although these mechanisms for causing mixing under ice are well known and many measurements of thermal structure in ice-covered lakes have been made, little theoretical work has been undertaken to provide a quantitative description of the physical processes.

The present aim is to construct simple mathematical models to quantify the influence of solar heating on convective mixing under ice in lakes. The models are tested using data from a number of lakes where the necessary physical parameters are known or can reasonably be assumed. An attempt is made to derive an approximate value for the vertical velocities of the water movements. The implications of the results are considered in relation to qualitative and quantitative aspects of biological production in ice-covered lakes.

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Density properties of water

To analyse mixing caused by an unstable density stratification, it is important to consider first the factors which control water density. The three main such factors are (a) temperature, (b) pressure, and (c) dissolved salts.

(a) Water has the property that its density ρ reaches a maximum ρ_4 at 4°C. Near this maximum, the density may be approximated by a quadratic function of temperature,

$$\rho(T)-\rho_4=-\alpha\rho_4(T-4)^2 \tag{1}$$

where $\rho_4=1000\text{ kg m}^{-3}$, $\alpha=8\times10^{-6}\text{ }^{\circ}\text{C}^{-2}$ and T is the temperature in $^{\circ}\text{C}$. Eqn (1) gives the density difference from ρ_4 with an error of less than 4% over the range 0–8°C.

(b) Although water is often assumed to be incompressible, the density does increase with hydrostatic pressure. The isothermal compressibility of water is $5.1\times10^{-10}\text{ Pa}^{-1}$ at 0°C (Eisenberg & Kauzmann, 1969), which means that the density increases by $4.9\times10^{-3}\text{ kg m}^{-3}$ for each metre of depth, at constant temperature. However, as a blob of water rises, the contribution to its density from depth changes as it does so to match that of its surroundings. Therefore the increase in density with depth does not affect stability and is not included in the models.

(c) If s is the ratio by weight of salt to water, the density of the solution is $1+0.8s$ times the density of pure water (Dorsey, 1940). For example, for the lakes of the English Lake District, s is around 5×10^{-5} , so a 20% increase in salt concentration produces a density increase of $8\times10^{-3}\text{ kg m}^{-3}$, which is equivalent to that due to a change in temperature from 3 to 4°C. Variation in salt concentration is, however, not taken into account in the remainder of this paper.

Solar heating

Provided that a significant amount of solar radiation penetrates the ice, the upper layers of an ice-covered lake are warmed. If the water temperature is below 4°C, the density of the warmed layers increases and an instability may develop, followed by convective mixing as the warmed water sinks. Whether or not this occurs depends on the initial temperature

TABLE 1. Mean total solar radiation for four localities (W m^{-2})

Site	Source	Lat.	Month												Yearly average
			J	F	M	A	M	J	J	A	S	O	N	D	
Karlsruhe, West Germany	Sauberer & Ruttner (1941)	49°N	33	72	116	166	243	251	254	179	146	98	33	24	135
English Lake District	Day (1961)	54°N	20	42	79	135	167	187	156	133	92	50	25	12	92
Signy Island, Maritime Antarctic	Walton (1977)	61°S	163	119	52	30	12	6	9	28	87	135	210	197	87
Great Bear Lake, Canada	Johnson (1975)	66°N	5	33	104	182	242	270	240	172	98	36	8	2	116

profile, the radiation intensity and the extinction coefficient.

The way in which solar radiation penetrates the water of an ice-covered lake may be represented by the equation

$$I(Z) = I_0(1-a)\exp(-\varepsilon_1 d - \varepsilon_2 Z) \quad (2)$$

where $I(Z)$ is the radiation intensity at a depth Z below the ice, I_0 is the radiation intensity at the upper surface of the ice, a is the albedo of the ice, d is the ice thickness and ε_1 and ε_2 are the vertical extinction coefficients of ice and water respectively.

I_0 is the total radiation received on a horizontal surface, including both direct solar radiation and diffuse radiation from the rest of the sky. Table 1 shows monthly averages for I_0 for four different localities. The polar regions show greater variation between summer and winter, but the annual average does not simply decrease with increasing latitude, due to variation in climate. Throughout this work I_0 is assumed to be constant for the period of sub-ice heating—daily and seasonal variations are not taken into account.

The albedo is the ratio of reflected to incident radiation. It increases from 0.25 to 0.65 for wet ice as the solar angle decreases (Adams, 1978) and also varies with the crystal structure of the ice, ranging from 0.2 to 0.35 for the ice cover of Arctic lakes (Hobbie, 1973). Maguire (1975) gave a value of 0.81 to 0.88 for fresh snow.

The extinction coefficient for ice varies

greatly with the quality of the ice. Clear colourless ice can transmit light as well as pure water (Sauberer, 1950), but bubbly white ice can have an extinction coefficient two orders of magnitude greater (Table 2). For both ice and water, the extinction coefficient depends critically on wavelength. For pure water, it is 0.6 m^{-1} at a wavelength of 700 nm and 0.013 m^{-1} at 400 nm (Hutchinson, 1957). Changing spectral composition with depth means that the use of a single extinction coefficient for total radiation is not strictly valid (e.g. Jewson *et al.*, 1984). A single extinction coefficient is assumed in the present work for simplicity and because of limited availability of data. The extinction coefficient varies with the concentration of biotic and abiotic components of the lake water. (Table 3 gives extinction coefficients and other physical data for lakes referred to in the text.)

Two computer models for studying the effect of solar heating on temperature and stability of ice-covered lakes were set up, and are described below.

(i) A one-dimensional lake model for solar heating

The radiation intensity at the ice–water interface under ice of thickness d is

$$I_1 = I_0(1-a)\exp(-\varepsilon_1 d) \quad (3)$$

At a depth Z below the ice it is

$$I(Z) = I_1 \exp(-\varepsilon_2 Z) \quad (4)$$

TABLE 2. Extinction coefficients for ice and snow for visible light (400–700 nm)

Source	Description	Spectral band of wavelength (nm)	$\varepsilon_1 \text{ (m}^{-1}\text{)}$
Ragotzkie & Likens (1964)	Clear ice, Antarctica	Visible	0.4–0.7
Hobbie (1973)	Clear ice, Arctic	Visible	0.2
	Bubbly ice	Visible	3.5
Maguire (1975)	Clear ice	Visible	2
	Cloudy ice	Visible	20
	Soft new snow	Visible	10
	Hard powder snow	Visible	50
Adams (1978)	White ice	Visible	4
	Snow	Visible	40
Sauberer (1950)	Clear ice, alpine	400	0.04
		500	0.08
		600	0.21
		700	0.60

TABLE 3. Physical data for lakes referred to in the text

Lake	Reference	Latitude	Area (ha)	Max. depth (m)	Minimum extinction coefficient (m ⁻¹)	Ice cover Duration (months)	Max. (m)
Heywood, Maritime Antarctica	Hawes (1983)	61°S	4.5	6.4	0.5	9	1.0
Sombre, Maritime Antarctica		61°S	2.7	11.2	0.2	9	1.0
Esthwaite Water, England	{Mortimer (1941) This work	54°N	100	15.5	0.4	up to 2	0.2
Babine, Canada		55°N	30,000	200	0.56	3	0.4
Bonney, Antarctica	Ragotzkie & Likens (1964)	78°S	408	32	0.07	Permanent	4.3
Ellesmere Island, Arctic Canada	Hattersley-Smith <i>et al.</i> (1970)	83°N	490	57	—	Permanent	2

from eqn (2). The rate of heat input to a layer of thickness dZ , assuming that the net incoming radiation is all converted into heat, is $I_1 \varepsilon_2 dZ \exp(-\varepsilon_2 Z)$, and the rate of temperature rise, assuming the water to be at rest, is

$$\frac{\partial T}{\partial t} = \frac{I_1 \varepsilon_2}{\rho C_p} \exp(-\varepsilon_2 Z) + K \frac{\partial^2 T}{\partial Z^2} \tag{5}$$

as given by Farmer (1975). C_p is the specific heat capacity of water, $4.2 \times 10^3 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$, and K is the vertical diffusivity, which has been determined as approximately $1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ by both Mortimer (1941), under ice in Esthwaite Water, English Lake District, and for two Antarctic lakes (British Antarctic Survey, unpublished data).

To find a unique solution to eqn (5), for the temperature as a function of depth and time, the initial temperature profile and two boundary conditions must be specified. For the initial temperature, a linear profile was used: thus at time $t=0$, the temperature is $T=A+BZ$, where A and B are constants. Two possible situations are envisaged which might lead to such a profile. For lakes in temperate regions, freezing usually occurs when the lake is well mixed. Taking this time as $t=0$, the temperature profile is then of the above form with $B=0$. For polar regions, there is a long winter period in which heating is negligible because of low I_0 values and snow cover. During this period, diffusion tends to smooth out the profile to a linear form. In this case, $t=0$ represents the time in spring when the snow melts and radiation begins to enter the lake. In each case the initial state is one of no motion except for the weak, small-scale eddies which cause the eddy diffusivity. The boundary conditions are that $T=0$ at the ice-water interface, $Z=0$, and that at large values of Z the temperature remains unchanged at its initial value of $A+BZ$ for all t . This means that the model is only applicable to a lake which is much deeper than its extinction path length, ε_2^{-1} . The form of the solution and its derivation are given in the appendix.

A computer program was written to plot profiles of temperature against depth, together with the corresponding density profiles calculated from temperature using eqn (1), at intervals of 4 days from 0 to 20 days. The program requires the parameters A , B , D (depth to which profiles are plotted), I_1 , K and

ϵ_2 . The temperature profiles can be compared with those of ice-covered lakes which are not mixed. The density profiles indicate whether or not mixing will occur due to solar heating, and if so, when the mixing begins and the approximate extent of the mixed layer. If the density profile shows a region where the density decreases with depth (a density inversion), then the water column becomes unstable and a mixed layer can be expected to form. The depth of the mixed layer may be estimated by drawing a line vertically downwards on the density profile, straightening its unstable part, as indicated in Fig. 1(b).

(ii) A one-dimensional layered model

The solution to eqn (5), used in model (i), assumes that temperature is the important boundary condition and that the temperature of the deep water remains constant in time. To allow a heat flux boundary condition, both at the ice surface and at the sediments, and to exhibit mixing, a layered computational model was used. The lake is represented by an arbitrary number of layers, the top layer being the ice cover, and each layer is given an initial temperature. The same physical parameters are required as for model (i), together with the heat flux through the ice and through the sediments. Using a time step of 8 h, the program calculates the new temperature of each layer due to radiation and diffusion, and corresponding density. If a layer is denser than one or more layers below, these layers are mixed to a uniform temperature and the densities are recalculated. The program also allows for melting of the ice if the temperature of the ice layer rises above 0°C , and thickening of the ice layer if the temperature of the first layer of water falls below 0°C . At the end of each day (three time steps) the program prints the temperature and density of each layer, the ice thickness and which layers if any are mixed. It was checked that altering the number of layers or the time step used did not significantly alter the results.

Results

Sensitivity of models to variation of parameters

The models were first used to establish the importance of each parameter for the forma-

tion of temperature and density profiles and the onset of mixing. The results of models (i) and (ii) are summarized in Tables 4 and 5 respectively. In most cases the results of the two models were in qualitative agreement.

Fig. 1(a) shows developing temperature and density profiles at 4-day intervals from 0 to 20 days as predicted by model (i), for an initial surface temperature of 1°C with a temperature gradient of 0.2°C m^{-1} . K is set to $1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$, the radiation intensity reaching the water is 15 W m^{-2} and ϵ_2 is 0.6 m^{-1} . It can be seen that the density profile remains stable. Figs. 1(b)–1(f) show the effect of altering each parameter in turn (except for D , which merely determines how much of the profile is plotted) and can be compared with Fig. 1(a).

Temperature (Fig. 1b). At a lower initial surface temperature of 0°C , the density profile just becomes unstable after 8 days and vertical mixing will take place after this time. This happens because the lower initial temperature reduces the stabilizing effect of the upper boundary condition, and because at a lower temperature a given temperature rise produces a greater density increase. The vertical line drawn on the density profile indicates how the mixed layer depth can be estimated.

Temperature gradient (Fig. 1c). If the initial temperature gradient is reduced to 0.1°C m^{-1} , but other parameters are as in Fig. 1(a), the water column becomes unstable after 8 days.

Diffusivity (Fig. 1d). If the diffusivity is reduced by a factor of 2, mixing occurs, because the temperature rise caused by the solar heating cannot spread out so rapidly.

Radiation intensity (Fig. 1e). If the incoming radiation is increased, the temperature rise of the water is greater and mixing occurs.

Extinction coefficient (Fig. 1f). With a higher extinction coefficient, mixing occurs, because the incoming radiation is absorbed by a thinner layer of water near the ice surface.

The Rayleigh number and the onset of convection

It is assumed throughout that mixing begins as soon as the density profile shows an inversion. In fact, very small density inversions are stable, and only become unstable when the Rayleigh number exceeds a certain critical

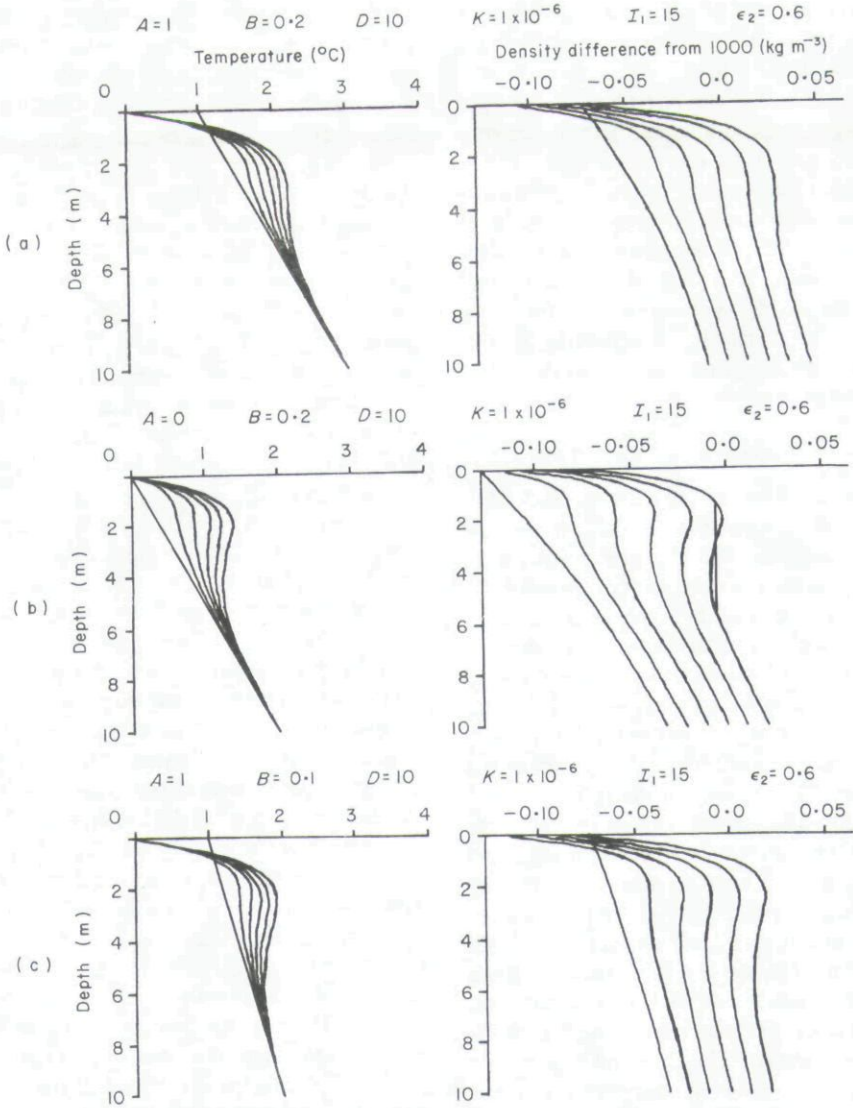


FIG. 1. Temperature and density profiles predicted by model (i), showing the dependence on: (a), (b), (c) initial temperature, $A+BZ$; (d) vertical diffusivity, K ; (e) solar radiation at ice–water interface, I_1 ; (f) extinction coefficient, ϵ_2 . D = depth to which profiles are plotted. Profiles are shown at 4-day intervals from 0 to 20 days. Density profiles are spaced for greater clarity.

value. The Rayleigh number is defined by

$$Ra = \frac{g \Delta \rho L^3}{\rho K \nu} \tag{6}$$

where g is the acceleration due to gravity, $\Delta \rho$ is the density difference which occurs over a vertical distance L and ν is the kinematic viscosity of water. Physically, the Rayleigh number represents the ratio of the destabilizing effect of the density inversion to the

stabilizing effects of viscosity and diffusion. A stability analysis of a density profile similar to those in Fig. 1 has shown that the critical Rayleigh number is around 100. When estimated from the density profiles in Fig. 1 showing a slight density inversion, however, Ra has a value of at least 10^6 . This is much greater than the critical value for the onset of convection. Hence the above assumption is justified. The stability analysis also showed

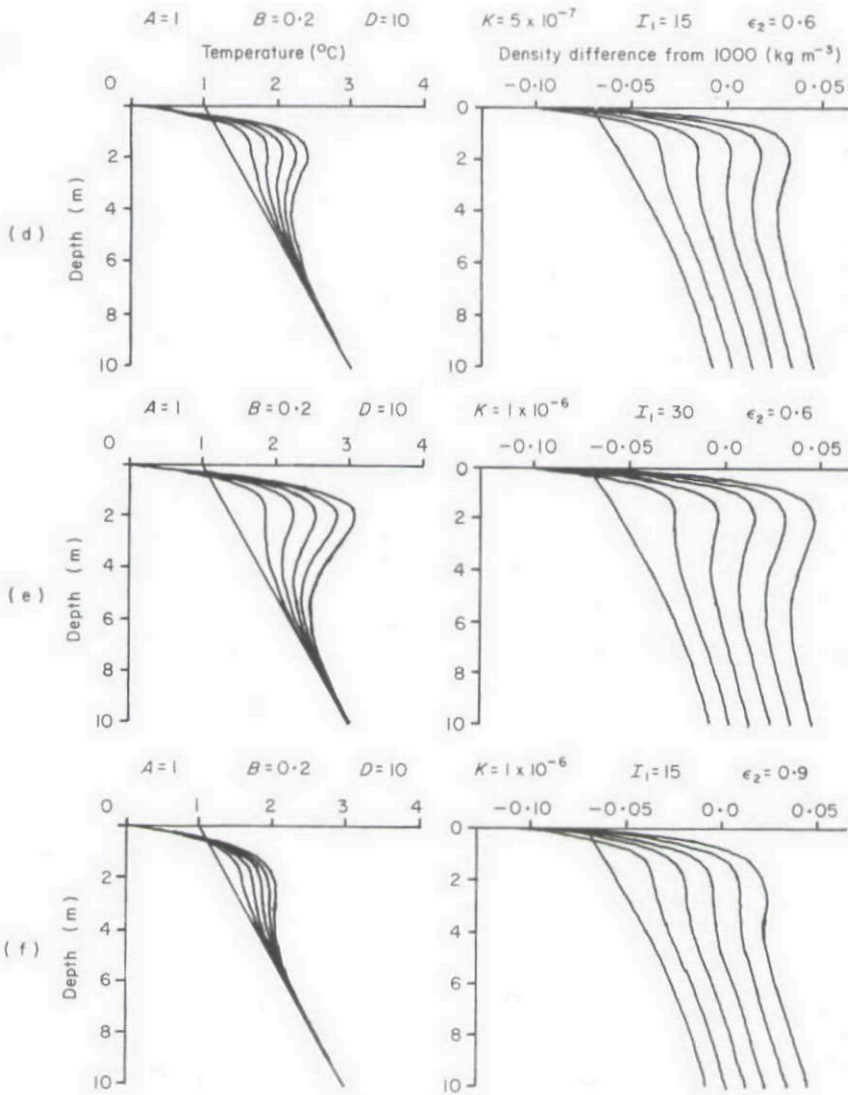


FIG. 1 (continued)

that the convection penetrates considerably into the unstable layer, so the mixed layer depth is underestimated by the simple method shown in Fig. 1(b).

Application of models to natural lakes

The models were applied to ice-covered lakes, where the necessary physical data are available, to assess whether the observed temperature profiles and mixing could be caused by solar heating. Table 3 gives some physical data for these lakes.

Babine Lake, British Columbia

The observations of Farmer (1975) showed a thickening mixed layer in Babine Lake during March and April 1973, a process known as penetrative convection. The mixed layer depth increased from 0 to 12 m in 22 days, and then rose to 28 m after a further 16 days. The temperature profile at the beginning of March was approximately linear in the upper 30 m, with a gradient of 0.1°C m^{-1} . The ice transmitted about one third of the incident radiation, so a value of 50 W m^{-2} was chosen for I_1 , for

TABLE 4. Summary of results of model (i)

Reference in text	Surface temperature (°C)	Temperature gradient (°C m ⁻¹)	Radiation penetrating ice (W m ⁻²)	Extinction coefficient (m ⁻¹)	Mixing	Time for onset of mixing (days)	Extent of mixed layer after 20 days (m)
Fig. 1(a)	1	0.2	15	0.6	No	—	—
Fig. 1(b)	0	0.2	15	0.6	Yes	8	2-6
Fig. 1(c)	1	0.1	15	0.6	Yes	8	2-8
Fig. 1(e)	1	0.2	30	0.6	Yes	8	1-8
Fig. 1(f)	1	0.2	15	0.9	Yes	20	2-4
Babine Lake, 1973	0	0.1	50	0.56	Yes	4	1-16
	0	0.1	30	0.56	Yes	4	1-13
Heywood Lake, 1973	1.5	0.25	30	0.8	Yes	8	1-6
	1.5	0.25	15	0.8	No	—	—
Heywood Lake, 1976	0.4	0.1	15	0.8	Yes	4	1-6
	0.4	0.1	10	0.8	Yes	8	1-6
Sombre Lake, 1976	0	0.15	15	0.25	No	—	—
	0	0.15	30	0.25	Yes	12	2-9

TABLE 5. Summary of results of model (ii)

Reference in text	Surface temperature (°C)	Temperature gradient (°C m ⁻¹)	Radiation penetrating ice (W m ⁻²)	Extinction coefficient (m ⁻¹)	Mixing	Time for onset of mixing (days)	Extent of mixed layer after 20 days (m)
Fig. 1(a)	1	0.2	15	0.6	No	—	—
Fig. 1(b)	0	0.2	15	0.6	Yes	12	2-5
Fig. 1(c)	1	0.1	15	0.6	Yes	5	3-8
Fig. 1(e)	1	0.2	30	0.6	Yes	4	2-9
Fig. 1(f)	1	0.2	15	0.9	Yes	16	3-4
Babine Lake, 1973	0	0.1	50	0.56	Yes	2	1-15
	0	0.1	30	0.56	Yes	3	1-12
Heywood Lake, 1973	1.5	0.25	30	0.8	Yes	5	2-6
	1.5	0.25	15	0.8	No	—	—
Heywood Lake, 1976	0.4	0.1	15	0.8	Yes	4	2-6
	0.4	0.1	10	0.8	Yes	8	2-6
Sombre Lake, 1976	0	0.15	15	0.25	No	—	—
	0	0.15	30	0.25	Yes	13	3-7

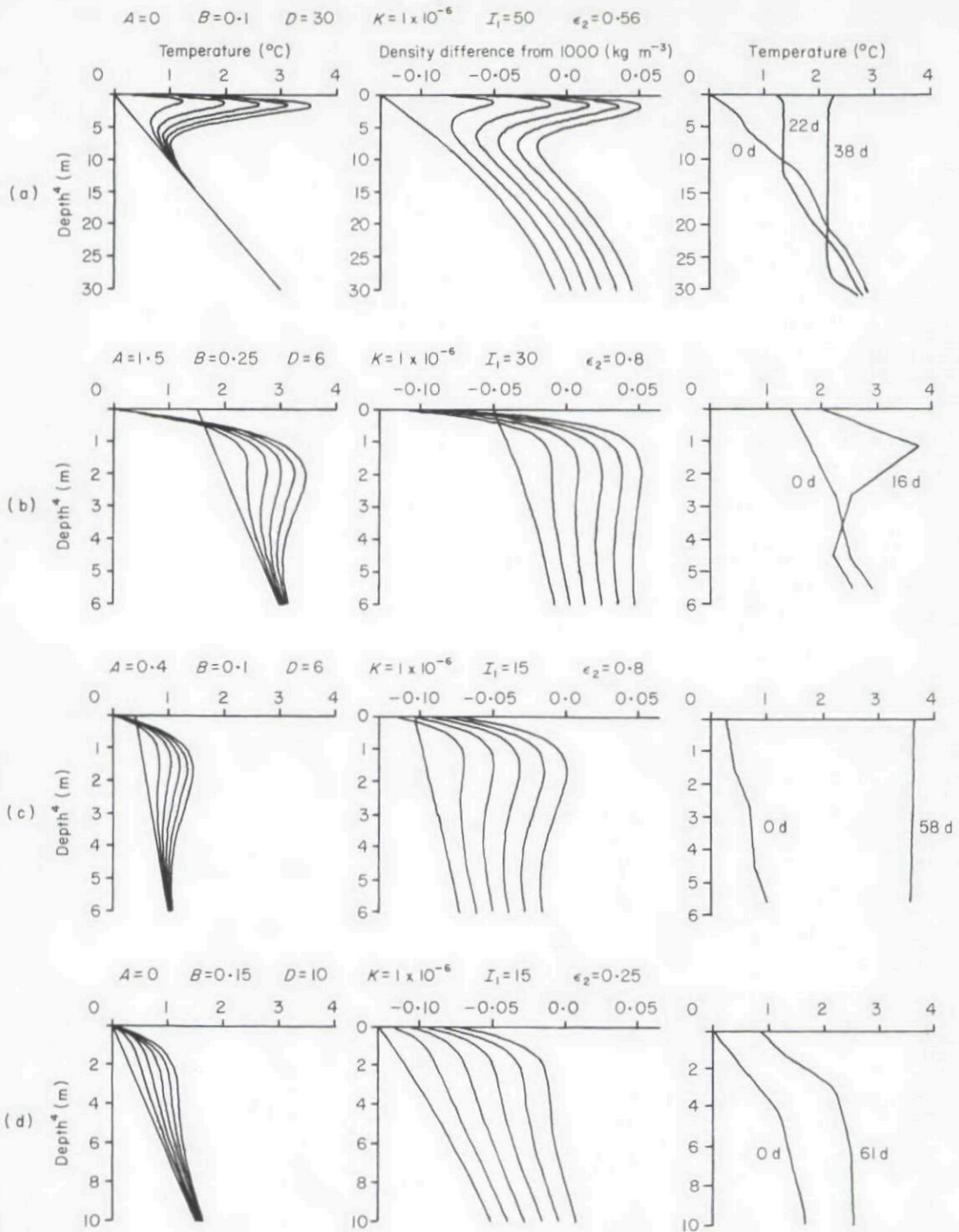


FIG. 2. Temperature and density profiles predicted by model (i), with observed temperature profiles: (a) Babine Lake, 1973; (b) Heywood Lake, 1973; (c) Heywood Lake, 1976; (d) Sombre Lake, 1976.

models (i) and (ii). A similar value was found by Farmer (1975) by considering the rate of temperature increase of the water column. A vertical diffusivity of $1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ was assumed. The extinction coefficient (wavelength not stated) for water below the ice was given as 0.56 m^{-1} . Fig. 2(a) shows the results of model (i) using these data together with the observed temperature profiles. Model (i) shows that an instability soon develops and the depth of the mixed layer increases to approximately 10 m after 8 days and 16 m after 20 days. Model (ii) gave similar results, showing mixing down to 10 m after 8 days and 15 m after 20 days (Tables 4 and 5).

Two maritime Antarctic lakes

Hawes (1983) made a comparative study of two small lakes on Signy Island which are ice-covered for most of the year. Measurements of temperature, chloride and chlorophyll-*a* during spring 1976 (October to December) indicated that there was vertical mixing in Heywood Lake, while the more oligotrophic Sombre Lake remained stratified. In spring 1973, however, mixing did not occur in Heywood Lake.

The vertical extinction coefficients for the two lakes for photosynthetically available radiation (PAR 400–700 nm) are given by Hawes (1985) as 0.8 m^{-1} for Heywood Lake and 0.25 m^{-1} for Sombre Lake. The value for Heywood Lake varies considerably during the spring period of phytoplankton growth. The radiation intensity below the ice in the absence of snow, also for PAR, is around 10 W m^{-2} in October, increasing to 20 W m^{-2} in December. However, we require the total radiation. To use the fact that PAR forms about half of total radiation would overestimate I_1 , because the extinction coefficient of ice is higher for infra-red light than for PAR. Using Table 1 and Hawes's value of around 15% for light penetration, it was assumed that I_1 increases from 15 to 30 W m^{-2} from October to December. K was set to $1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$, the value found for these lakes (British Antarctic Survey, unpublished data).

At the end of November 1973 the temperature profile of Heywood Lake was linear, 1.5°C at the surface and 3°C at the bottom (6 m). Both models predict mixing for these

initial conditions, using $I_1 = 30 \text{ W m}^{-2}$. Although the observations of Hawes (1983) indicated that the lake was not mixed, the temperature profiles predicted by model (i) agree with the observations (Fig. 2b, Tables 4 and 5). Since these temperature profiles are unstable, mixing was probably prevented by a stable salinity gradient. In October 1976, however, the initial temperature range was 0.4°C – 1.0°C , and for these values both models predict mixing, with the lower value of I_1 appropriate to October (Fig. 2c, Tables 4 and 5). The density gradient shown by model (i) for this case is considerably greater than that for 1973. For Sombre Lake, the temperature range was 0°C below the ice increasing to 1.5°C at 10 m, and using the lower value for I_1 of 15 W m^{-2} , mixing does not occur (Fig. 2d, Tables 4 and 5). In fact Sombre Lake is partially shaded by a hill to the north, which reduces I_1 and makes mixing less likely. Again the observed and predicted temperature profiles agree quite well.

Velocity estimates

Rough estimates of the vertical velocities involved in convection due to solar heating can be made from the results of model (ii). In the model, layers which are unstably stratified are mixed instantaneously to form a mixed layer which warms and deepens in time. In this mixed layer, eqn (5) must be modified to incorporate the water motion:

$$\frac{\partial T}{\partial t} + \frac{\partial}{\partial Z} \langle WT' \rangle = \frac{I_1 \epsilon_2}{\rho C_p} \exp(-\epsilon_2 Z) + K \frac{\partial^2 T}{\partial Z^2} \quad (7)$$

where $\langle \rangle$ denotes a horizontal average, W is the vertical velocity and T' is the temperature perturbation from its average value in the mixed layer, T . The last term in (7) is zero in the mixed layer because T is uniform there. Integrating this equation in the Z direction gives

$$\langle WT' \rangle = \frac{I_1}{\rho C_p} [\exp(-\epsilon_2 Z_0) - \exp(-\epsilon_2 Z)] - (Z - Z_0) \frac{\partial T}{\partial t} \quad (8)$$

where Z_0 is the depth of the top of the mixed layer. From this we can estimate W using the following formula for a fully turbulent convect-

ing layer, derived from dimensional analysis and supported by experiment (Townsend, 1976, p. 388):

$$W \approx (hg\alpha' \langle WT' \rangle)^{1/3} \quad (9)$$

where h is the mixed layer depth and α' is the thermal expansion coefficient.

For example, in the application of model (ii) to the spring 1976 mixing of Heywood Lake, the temperature of the mixed layer was around 1°C , increasing at a rate of 0.02°C per day, and the extent of the mixed layer was 4 m. Applying eqn (8) in the middle of the mixed layer gives $\langle WT' \rangle = 1.1 \times 10^{-7} \text{ m } ^\circ\text{C s}^{-1}$, so, from eqn (9), $W \approx 6.0 \times 10^{-4} \text{ m s}^{-1}$, which is 52 m day^{-1} . For the case of Babine Lake, the temperature was again around 1°C , rising at 0.05°C per day, and the mixed layer depth was 14 m after 20 days. These figures give $W \approx 2.6 \times 10^{-3} \text{ m s}^{-1}$, or 220 m day^{-1} . This result agrees well with the calculations made by Farmer (1975), using the difference in phase between temperature fluctuations at different depths, which gave a result of $2.2 \times 10^{-3} \text{ m s}^{-1}$.

Mixing and phytoplankton production

The scale of vertical velocities of water movements is an important variable in determining

phytoplankton production and species composition. For populations of non-motile species to increase, their sinking rates must be less than vertical water velocities due to convection or turbulence. All other requirements being equal, motile or buoyant species will be advantaged when their capacities for active depth regulation are greater than vertical water movements. The differential selection of non-motile diatoms under ice cover in relation to their sinking and growth characteristics has been observed by Lund (1959). However, if these characteristics are known, then there is a possibility of using changes of population dynamics to indicate the scale of vertical water movements when weak mixing occurs.

This approach is considered using diatom populations in Esthwaite Water, sampled from the 0–5 m layer with a plastic tube. Changes in population density of *Asterionella formosa* Hass. and *Melosira italica* subsp. *subarctica* O. F. Mull., counted using the inverted microscope technique (Lund, Kipling & Le Cren, 1958) are shown in Fig. 3 for two contrasting years. 1979 was a year with prolonged ice cover but with apparent mixing in the 1–5 m layer and 1981 an ice-free year with isothermal conditions during winter. Under ice cover the *Asterionella* population increased but that of *Melosira* did not, whereas during 1981 both

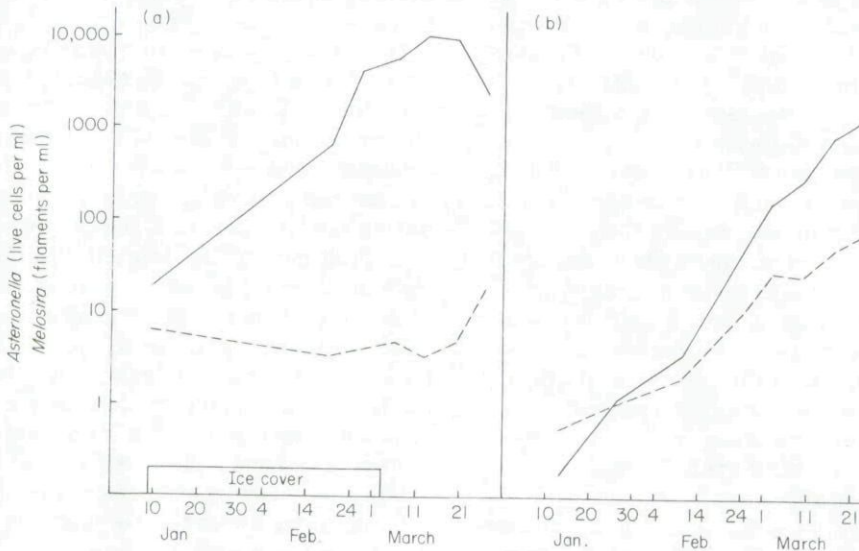


FIG. 3. Growth of *Asterionella formosa* (continuous line) and *Melosira italica* (dashed line) in Esthwaite Water in two contrasting years: (a) 1979, with ice cover; (b) 1981, without ice cover.

species increased, although the former more rapidly in accordance with its faster growth rate (Lund, 1959). Measurements of sinking rates of actively growing cells of *A. formosa* and *M. italica* subsp. *subarctica* are c. 0.2 and 1.0 m day⁻¹ respectively (Lund, 1959). Taken together, these observations suggest that time-averaged vertical velocities in the 1–5 m layer below the ice in 1979 would have been between the sinking rates of the two species.

Discussion

This study shows that solar heating can be important in determining temperature and hence density in ice-covered lakes, and can provide the driving energy for penetrative convection (Farmer, 1975; Petrov, 1984) provided that certain conditions are satisfied. The factors which make a lake likely to mix due to solar heating are (a) high solar radiation input, (b) high extinction coefficient, (c) low initial temperature, (d) low density stratification due to temperature gradient or dissolved substances, and (e) low vertical eddy diffusivity.

The need for high radiation input means that solar heating can be important throughout the winter in temperate regions (dependent on cloud cover) and during spring in polar areas, in lakes without snow cover. Owing to its high albedo and extinction coefficient, 0.1 m of snow can reduce incident radiation by a factor of 100. Solar heating is likely to be more significant in ice-covered eutrophic lakes or highly coloured dystrophic lakes, both with high extinction coefficients.

Of the two Signy Island lakes studied by Hawes (1983), Sombre Lake is less susceptible to mixing due to solar heating than Heywood Lake, because of its lower extinction coefficient and partial shading. In spring 1973, Heywood Lake did not mix despite its unstable temperature profile, probably because of salt stratification. In 1976, however, mixing did occur in Heywood Lake because the temperature and temperature gradient were lower, while Sombre Lake remained stratified for the reasons given above.

Further evidence that solar heating is important in ice-covered lakes comes from the temperature profiles of lakes with stabilizing salinity gradients. Hattersley-Smith *et al.*

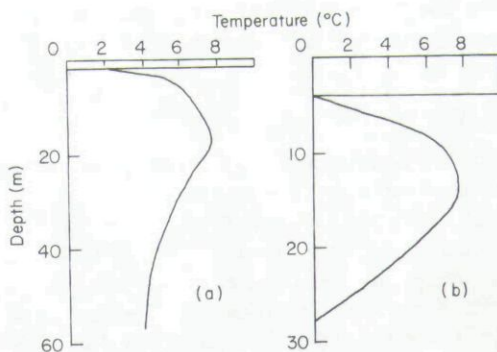


FIG. 4. Temperature profiles of two ice-covered lakes which are stabilized by a salinity gradient: (a) a lake on Ellesmere Island (Hattersley-Smith *et al.*, 1970); (b) Lake Bonney, Antarctica (Ragotzkie & Likens, 1964).

(1970) observed temperature profiles in lakes with a salinity gradient under 2 m of ice at latitude 83°N during May, and attributed their shape to solar heating (Fig. 4a). Similarly shaped profiles of temperature were found in Lake Bonney in Antarctica (Fig. 4b) which has a permanent ice cover of 4 m thickness (Ragotzkie & Likens, 1964). Model (i) cannot be directly applied to these lakes because data on time evolution of the profiles and light penetration are not available, but qualitatively the profiles are similar to those in Fig. 1. Shirtcliffe & Benseman (1965) successfully fitted the temperature profile of Lake Bonney to a steady-state solution of eqn (5).

The mixing velocities suggested by the results of model (ii) are considerably greater than the sinking rates of most freshwater phytoplankton. This is important in helping to determine which species will occur. In ice-covered lakes which are stratified, flagellates usually tend to dominate the phytoplankton (e.g. Pechlaner, 1971; Hobbie, 1973; Hawes, 1983; Rott, 1983) as they can take advantage of depth regulation for positioning in light and nutrient gradients. For those lakes in which mixing takes place under ice, by whatever mechanism, non-motile species may remain in suspension and grow, e.g. Heywood Lake (Hawes, 1983) and Lake Baikal (Lund, 1966).

Physical prediction of whether or not mixing will take place under ice can give a guide to qualitative and quantitative biological and chemical changes which can be expected. A greater understanding of the growth and sinking

rates of phytoplankton species may in turn give an independent check on determinations of the magnitudes of water movements.

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References

- Adams W.A. (1978) Effects of ice cover on the solar radiation regime in Canadian lakes. *Verhandlungen der Internationalen Vereinigung für Theoretische und Angewandte Limnologie*, **20**, 141–149.
- Bergmann M.A. & Welch H.E. (1985) Spring melt-water mixing in small Arctic lakes. *Canadian Journal of Fisheries and Aquatic Sciences*, **42**, 1789–1798.
- Day G.J. (1961) Distribution of solar radiation over the British Isles. *Meteorological Magazine*, **90**, 269–284.
- Dorsey N.E. (1940) *Properties of Ordinary Water-Substance*. Reinhold, New York.
- Eisenberg D. & Kauzmann W. (1969) *The Structure and Properties of Water*. Clarendon Press, Oxford.
- Farmer D.M. (1975) Penetrative convection in the absence of mean shear. *Quarterly Journal of the Royal Meteorological Society*, **101**, 869–891.
- Hattersley-Smith G., Keys J.E., Serson H. & Mielke J.E. (1970) Density stratified lakes in Northern Ellesmere Island. *Nature*, **225**, 55–56.
- Hawes I. (1983) Turbulence and its consequences for phytoplankton development in two ice-covered Antarctic lakes. *British Antarctic Survey Bulletin*, **60**, 69–81.
- Hawes I. (1985) Light climate and phytoplankton photosynthesis in maritime Antarctic lakes. *Hydrobiologia*, **123**, 69–79.
- Hobbie J.E. (1973) Arctic limnology: a review. *Alaskan Arctic Tundra* (Ed. M. E. Britton), pp. 127–168. Arctic Institute of North America Technical paper, No. 25.
- Hutchinson G.E. (1957) *A Treatise on Limnology*, Vol. 1. John Wiley and Sons, New York.
- Jewson D.H., Talling J.F., Dring M.J., Tilzer M.M., Heaney S.I. & Cunningham C. (1984) Measurement of photosynthetically available radiation in freshwater: comparative tests of some current instruments used in studies of primary production. *Journal of Plankton Research*, **6**, (2), 259–273.
- Johnson L. (1975) Physical and chemical characteristics of Great Bear Lake. *Journal of the Fisheries Research Board of Canada*, **32**, 1971–1987.
- Likens G.E. & Ragotzkie R.A. (1965) Vertical water motions in a small ice-covered lake. *Journal of Geophysical Research*, **70**, 2333–2344.
- Likens G.E. & Ragotzkie R.A. (1966) Rotary circulation of water in an ice-covered lake. *Verhandlungen der Internationalen Vereinigung für Theoretische und Angewandte Limnologie*, **16**, 126–133.
- Lund J.W.G. (1959) Buoyancy in relation to the ecology of the freshwater phytoplankton. *British Phycological Bulletin*, **1**, 1–17.
- Lund J.W.G. (1966) The importance of turbulence in the periodicity of certain freshwater species of the genus *Melosira*. *Botanicheskii Zhurnal*, **51**, 176–187.
- Lund J.W.G., Kipling C. & Le Cren E.D. (1958) The inverted microscope method of estimating algal numbers and the statistical basis of estimations by counting. *Hydrobiologia*, **11**, 143–170.
- Maguire R.J. (1975) *Effects of Ice and Snow Cover on Transmission of Light in Lakes*. Environment Canada, Burlington, Ontario.
- Mortimer C.H. (1941) The exchange of dissolved substances between mud and water in lakes. *Journal of Ecology*, **29**, 280–329.
- Pechlaner R. (1971) Factors that control production rates and biomass of phytoplankton in high mountain lakes. *Verhandlungen der Internationalen Vereinigung für Theoretische und Angewandte Limnologie*, **19**, 125–145.
- Petrov M.P. (1984) Penetrative convection in a period of heating under ice in Lake Onezsk. *Hydrology of Lake Baikal and Other Water Bodies* (Ed. V. I. Verbovolov), pp. 88–92. Novosibirsk, Nauka.
- Ragotzkie R.A. & Likens G.E. (1964) The heat balance of two Antarctic lakes. *Limnology and Oceanography*, **9**, 412–425.
- Rott E. (1983) Are changes in phytoplankton assemblages of Piburger See consequences of hypolimnetic water removal? *Archiv für Hydrobiologie*, **67**, (1), 29–80.
- Sauberer F. (1950) Die spektrale Strahlung durchlässigkeit des Eises. *Wetter und Leben*, **2**, 193–197.
- Sauberer F. & Ruttner F. (1941) *Die Strahlungsverhältnisse der Binnengewässer*. Akademische Verlagsgesellschaft, Leipzig.
- Shirtcliffe T.G.L. & Benseman R.F. (1965) A sun heated Antarctic lake. *Journal of Geophysical Research*, **69**, 3355–3359.
- Sokolnikov V.M. (1960) On currents and water temperature under ice cover in the southern region of Lake Baikal and around the outflow of the Angar river. *Trudy Baikalskoi Limnologicheskoi Stantsii*, **18**, 264–265.
- Stewart K.M. (1973) Winter conditions in Lake Erie with reference to ice and thermal structure and comparison to lakes Winnebago (Wisconsin) and Mille lacs (Minnesota). *Proceedings of the 16th Conference of Great Lakes Research*, pp. 845–857.

- Townsend A.A. (1976) *The Structure of Turbulent Shear Flow*. Cambridge University Press.
- Walton D.W.H. (1977) Radiation and soil temperatures 1972–1974: Signy Island Terrestrial Reference Site. *British Antarctic Survey Data*, **1**, 1–50.
- Welch H.E. & Bergmann M.A. (1985) Water circulation in small Arctic lakes in winter. *Canadian Journal of Fisheries and Aquatic Sciences*, **42**, 506–520.
- Woodcock A.H. & Riley G.A. (1947) Patterns in pond ice. *Journal of Meteorology*, **4**, 100–101.

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List of symbols

Symbol	Meaning	Units
A	initial surface temperature	$^{\circ}\text{C}$
a	albedo	—
B	initial temperature gradient	$^{\circ}\text{C m}^{-1}$
C_p	specific heat capacity of water	$\text{J kg}^{-1} ^{\circ}\text{C}^{-1}$
D	lake depth (model (i))	m
d	ice thickness	m
g	acceleration due to gravity	m s^{-2}
h	layer thickness (model (ii))	m
I_0	surface radiation intensity	W m^{-2}
I_1	radiation intensity below ice	W m^{-2}
K	vertical eddy diffusivity	$\text{m}^2 \text{s}^{-1}$
L	vertical height of density inversion	m
n	number of layers (model (ii))	—
Ra	Rayleigh number	—
s	ratio of solutes to water by weight	—
T	temperature	$^{\circ}\text{C}$
t	time	s
W	vertical fluid velocity	m d^{-1}

Z	depth below ice-water surface	m
Z_0	top of mixed layer	m
α	constant $= 8 \times 10^{-6}$	$^{\circ}\text{C}^{-2}$
α'	coefficient of thermal expansion	$^{\circ}\text{C}^{-1}$
ε_1	extinction coefficient for ice	m^{-1}
ε_2	extinction coefficient for water	m^{-1}
ν	kinematic viscosity of water	$\text{m}^2 \text{s}^{-1}$
ρ	density of water	kg m^{-3}

Appendix

Derivation of solution for model (i)

Starting from eqn (5), we have

$$\frac{\partial T}{\partial t} = C \exp(-\varepsilon_2 Z) + K \frac{\partial^2 T}{\partial Z^2} \quad (5a)$$

where $C = I_1 \varepsilon_2 / \rho C_p$.

Take Laplace transform in time:

$$p \tilde{T}(Z, p) - A - BZ = \frac{C}{p} \exp(-\varepsilon_2 Z) + K \frac{\partial^2 \tilde{T}}{\partial Z^2}(Z, p) \quad (6a)$$

where p is the Laplace transform variable and $A + BZ$ is the initial temperature profile.

The boundary conditions are:

- (i) $Z=0$: $T=0$.
- (ii) $Z \rightarrow \infty$: $T = A + BZ$.

Seek a solution of the form

$$\tilde{T} = c_1 \exp[\sqrt{(p/K)}Z] + c_2 \exp[-\sqrt{(p/K)}Z] + \alpha \exp(-\varepsilon_2 Z) + \frac{A + BZ}{p} \quad (7a)$$

where α , c_1 and c_2 are constants.

Substituting (7a) into (6a), we find

$$\alpha = \frac{C}{p^2 - p\varepsilon_2^2 K} \quad (8a)$$

Using (ii), $c_1=0$.

To find c_2 , set $Z=0$ in (7a) and use (i):

$$0=c_2+\alpha+\frac{A}{p} \quad (9a)$$

$$c_2=-\frac{A}{p}+\frac{C}{K\varepsilon_2^2p}-\frac{C}{K\varepsilon_2^2(p-K\varepsilon_2^2)} \quad (10a)$$

so we have the Laplace transform of T ,

$$\begin{aligned} \hat{T}(Z,p)= & \left(-A+\frac{C}{K\varepsilon_2^2}\right)\frac{\exp[-\sqrt{(p/K)}Z]}{p} \\ & -\frac{C\exp[-\sqrt{(p/K)}Z]}{K\varepsilon_2^2(p-K\varepsilon_2^2)} \\ & +\frac{C\exp(-\varepsilon_2Z)}{(p^2-p\varepsilon_2^2K)}+\frac{A+BZ}{p}. \end{aligned} \quad (11a)$$

To invert the Laplace transform, the first two terms are tabulated (e.g. in Carslow & Jaeger,

Conduction of Heat in Solids, Clarendon Press, 1947). The third term has a contribution $-(C/K\varepsilon_2^2)\exp(-\varepsilon_2Z)$ from the pole at zero, and $(C/K\varepsilon_2^2)\exp(-\varepsilon Z)\exp(K\varepsilon_2^2t)$ from the pole at $p=\varepsilon_2^2K$. The solution is:

$$\begin{aligned} T(Z,t)= & \left(\frac{C}{K\varepsilon_2^2}-A\right)\operatorname{erfc}\left(\frac{Z}{2\sqrt{Kt}}\right) \\ & -\frac{C}{2K\varepsilon_2^2}\exp(K\varepsilon_2^2t)\times \\ & \left[\exp(-\varepsilon_2Z)\operatorname{erfc}\left(\frac{Z}{2\sqrt{Kt}}-\varepsilon_2\sqrt{Kt}\right)\right. \\ & \left.+\exp(\varepsilon_2Z)\operatorname{erfc}\left(\frac{Z}{2\sqrt{Kt}}+\varepsilon_2\sqrt{Kt}\right)\right] \\ & +\frac{C}{K\varepsilon_2^2}\exp(-\varepsilon_2Z)[\exp(\varepsilon_2^2Kt)-1] \\ & +A+BZ. \end{aligned} \quad (12a)$$

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